

Homework 7 Solutions

BSTA 551

Problem 1: Two-Sample t-Tests for Means

Part (a) — Welch–Satterthwaite degrees of freedom

Summary statistics:

```
n1 <- 22; xbar1 <- 412.8; s1 <- 68.4 # COPD
n2 <- 25; xbar2 <- 356.2; s2 <- 41.7 # Controls

# Welch-Satterthwaite
v1 <- s1^2 / n1
v2 <- s2^2 / n2

nu_hat <- (v1 + v2)^2 / (v1^2 / (n1 - 1) + v2^2 / (n2 - 1))
cat(sprintf("v1 = s1^2/n1 = %.4f\n", v1))
```

$v1 = s1^2/n1 = 212.6618$

```
cat(sprintf("v2 = s2^2/n2 = %.4f\n", v2))
```

$v2 = s2^2/n2 = 69.5556$

```
cat(sprintf("Numerator: (v1 + v2)^2 = %.4f\n", (v1 + v2)^2))
```

Numerator: $(v1 + v2)^2 = 79646.6711$

```
cat(sprintf("Denominator: %.4f\n", v1^2/(n1-1) + v2^2/(n2-1)))
```

Denominator: 2355.1563

```
cat(sprintf("nu_hat = %.2f --> round down to %d\n", nu_hat, floor(nu_hat)))
```

nu_hat = 33.82 --> round down to 33

$$\hat{\nu} = \frac{(s_1^2/n_1 + s_2^2/n_2)^2}{\frac{(s_1^2/n_1)^2}{n_1 - 1} + \frac{(s_2^2/n_2)^2}{n_2 - 1}}$$

The Welch–Satterthwaite approximation gives $\hat{\nu} \approx 33.8$, rounded down to 33 degrees of freedom.

Part (b) — Welch t-test (one-sided)

```
SE_diff <- sqrt(s1^2/n1 + s2^2/n2)
t_obs   <- (xbar1 - xbar2) / SE_diff
p_one   <- pt(t_obs, df = nu_hat, lower.tail = FALSE)

cat(sprintf("SE(xbar1 - xbar2) = %.4f\n", SE_diff))
```

SE(xbar1 - xbar2) = 16.7993

```
cat(sprintf("t_obs = %.4f\n", t_obs))
```

t_obs = 3.3692

```
cat(sprintf("df      = %.2f\n", nu_hat))
```

df = 33.82

```
cat(sprintf("p-value (one-sided, upper) = %.5f\n", p_one))
```

p-value (one-sided, upper) = 0.00095

Test statistic: $T_W = \frac{412.8 - 356.2}{\sqrt{68.4^2/22 + 41.7^2/25}} = 3.369$

Degrees of freedom: $\hat{\nu} \approx 33.8$

p-value: 9.5×10^{-4}

Conclusion: At $\alpha = 0.05$ we reject H_0 ($p < 0.05$). There is statistically significant evidence that mean plasma fibrinogen is higher in COPD patients than in healthy controls.

Part (c) — 95% two-sided CI

```
t_crit <- qt(0.975, df = nu_hat)
ci_lo  <- (xbar1 - xbar2) - t_crit * SE_diff
ci_hi  <- (xbar1 - xbar2) + t_crit * SE_diff

cat(sprintf("t_crit (df = %.1f) = %.4f\n", nu_hat, t_crit))
```

t_crit (df = 33.8) = 2.0326

```
cat(sprintf("95%% CI: (%.2f, %.2f) mg/dL\n", ci_lo, ci_hi))
```

95% CI: (22.45, 90.75) mg/dL

Interpretation: We are 95% confident that the true mean plasma fibrinogen is between 22.5 and 90.7 mg/dL higher in COPD patients than in healthy controls. Because the entire interval is positive and excludes zero, the data provide strong evidence of a meaningful difference.

Part (d) — F-test for equality of variances

```

F_obs <- s1^2 / s2^2
df1    <- n1 - 1
df2    <- n2 - 1

# Two-sided p-value: 2 * min(lower, upper) tail
p_F <- 2 * min(pf(F_obs, df1, df2), pf(F_obs, df1, df2, lower.tail = FALSE))

cat(sprintf("F = s1^2/s2^2 = %.4f^2 / %.4f^2 = %.4f\n", s1, s2, F_obs))

```

$F = s_1^2/s_2^2 = 68.4000^2 / 41.7000^2 = 2.6905$

```
cat(sprintf("df1 = %d, df2 = %d\n", df1, df2))
```

df1 = 21, df2 = 24

```
cat(sprintf("Two-sided p-value = %.5f\n", p_F))
```

Two-sided p-value = 0.02115

Hypotheses: $H_0 : \sigma_1^2 = \sigma_2^2$ vs. $H_a : \sigma_1^2 \neq \sigma_2^2$

Test statistic: $F_0 = s_1^2/s_2^2 = 4678.6/1738.9 = 2.691$ on (21, 24) df

p-value: 0.0211

The F -test is highly significant, providing strong evidence that the population variances are *unequal*. **The Welch t -test is therefore the more appropriate choice** because it does not assume equal variances. The pooled t -test would be invalid here.

Part (e) — Pooled t -test for comparison

```

Sp2 <- ((n1-1)*s1^2 + (n2-1)*s2^2) / (n1 + n2 - 2)
Sp  <- sqrt(Sp2)
SE_pool <- Sp * sqrt(1/n1 + 1/n2)
t_pool <- (xbar1 - xbar2) / SE_pool
df_pool <- n1 + n2 - 2

```

```
p_pool <- pt(t_pool, df = df_pool, lower.tail = FALSE)
t_crit_pool <- qt(0.975, df = df_pool)
ci_pool <- c((xbar1 - xbar2) - t_crit_pool * SE_pool,
            (xbar1 - xbar2) + t_crit_pool * SE_pool)

cat(sprintf("Sp^2 = %.2f --> Sp = %.2f\n", Sp2, Sp))
```

Sp^2 = 3110.74 --> Sp = 55.77

```
cat(sprintf("SE_pool = %.4f\n", SE_pool))
```

SE_pool = 16.3042

```
cat(sprintf("t_pool = %.4f, df = %d\n", t_pool, df_pool))
```

t_pool = 3.4715, df = 45

```
cat(sprintf("p-value (one-sided) = %.5f\n", p_pool))
```

p-value (one-sided) = 0.00058

```
cat(sprintf("95%% Pooled CI: (%.2f, %.2f)\n", ci_pool[1], ci_pool[2]))
```

95% Pooled CI: (23.76, 89.44)

Comparison:

Method	t	df	95% CI	p -value
Welch	3.369	33.8	(22.5, 90.7)	9.5×10^{-4}
Pooled	3.472	45	(23.8, 89.4)	5.8×10^{-4}

The two procedures give similar conclusions here because both n_1 and n_2 are moderate. However, the pooled SE is inflated relative to the Welch SE because the larger variance (COPD, $s_1 = 68.4$) is averaged with the smaller (Controls, $s_2 = 41.7$). When sample sizes are **unequal and the larger variance is in the smaller group**, the pooled test can be anti-conservative (actual Type I error exceeds nominal α). The Welch test does not rely on equal-variance assumptions and is generally preferred.

Problem 2: Two-Sample Inference for Proportions

```
nA <- 105; xA <- 89 # Regimen A
nB <- 105; xB <- 76 # Regimen B

phatA <- xA / nA
phatB <- xB / nB
diff_hat <- phatA - phatB

cat(sprintf("phat_A = %d/%d = %.4f\n", xA, nA, phatA))
```

phat_A = 89/105 = 0.8476

```
cat(sprintf("phat_B = %d/%d = %.4f\n", xB, nB, phatB))
```

phat_B = 76/105 = 0.7238

```
cat(sprintf("phat_A - phat_B = %.4f\n", diff_hat))
```

phat_A - phat_B = 0.1238

Part (a) — Verify large-sample conditions

The four conditions for $\hat{p}_1 - \hat{p}_2 \approx N(p_1 - p_2, SE^2)$:

```
cond <- tibble(
  Regimen = c("A", "A", "B", "B"),
  Condition = c("n*phat", "n*(1-phat)", "n*phat", "n*(1-phat)"),
  Value = c(nA*phatA, nA*(1-phatA), nB*phatB, nB*(1-phatB))
)
cond
```

```
# A tibble: 4 x 3
  Regimen Condition Value
  <chr>    <chr>    <dbl>
1 A      n*phat      89
2 A      n*(1-phat)  16
3 B      n*phat      76
4 B      n*(1-phat)  29
```

All four values exceed 5 (in fact all exceed 10), so the normal approximation is appropriate.

Part (b) — 95% CI (unpooled SE)

```
SE_unpooled <- sqrt(phatA*(1-phatA)/nA + phatB*(1-phatB)/nB)
z_crit      <- qnorm(0.975)
ci_lo_p     <- diff_hat - z_crit * SE_unpooled
ci_hi_p     <- diff_hat + z_crit * SE_unpooled

cat(sprintf("SE (unpooled) = %.5f\n", SE_unpooled))
```

SE (unpooled) = 0.05598

```
cat(sprintf("95%% CI: (%.4f, %.4f)\n", ci_lo_p, ci_hi_p))
```

95% CI: (0.0141, 0.2335)

```
cat(sprintf("95%% CI: (%.1f%%, %.1f%%)\n", 100*ci_lo_p, 100*ci_hi_p))
```

95% CI: (1.4%, 23.4%)

Interpretation: We are 95% confident that the true success rate for Regimen A is between 1.4% and 23.4% higher than for Regimen B. Since the interval contains 0, the data do **not** provide statistically significant evidence of a difference at the 5% level (though the point estimate favors A).

Part (c) — Two-sided z-test (pooled SE)

```
phat_pool <- (xA + xB) / (nA + nB)
SE_pool_p <- sqrt(phat_pool*(1-phat_pool)*(1/nA + 1/nB))
Z_obs     <- diff_hat / SE_pool_p
p_two     <- 2 * pnorm(abs(Z_obs), lower.tail = FALSE)

cat(sprintf("phat_pool = (%d+%d)/(%d+%d) = %.4f\n", xA, xB, nA, nB, phat_pool))
```

phat_pool = (89+76)/(105+105) = 0.7857

```
cat(sprintf("SE (pooled) = %.5f\n", SE_pool_p))
```

SE (pooled) = 0.05663

```
cat(sprintf("Z = %.4f\n", Z_obs))
```

Z = 2.1863

```
cat(sprintf("p-value (two-sided) = %.5f\n", p_two))
```

p-value (two-sided) = 0.02880

```
# Verify
prop.test(c(xA, xB), c(nA, nB), correct = FALSE) |> tidy()
```

```
# A tibble: 1 x 9
  estimate1 estimate2 statistic p.value parameter conf.low conf.high method
    <dbl>     <dbl>     <dbl>  <dbl>     <dbl>     <dbl>     <dbl> <chr>
1    0.848     0.724     4.78  0.0288         1    0.0141    0.234 2-
sample t~
# i 1 more variable: alternative <chr>
```

Test statistic: $Z = 2.186$, $p\text{-value} = 0.0288$

At $\alpha = 0.05$ we fail to reject H_0 . There is insufficient evidence to conclude that the success rates differ between the two regimens.

Part (d) — Pooled vs. unpooled SE: conceptual explanation

When we test $H_0 : p_A = p_B$, we assume under H_0 that there is a **single common success probability** p . The pooled estimate $\hat{p}_{\text{pool}} = (x_A + x_B)/(n_A + n_B)$ is the most efficient estimate of this common p under H_0 , so plugging it into the SE is consistent with the null hypothesis and yields a valid test.

For the CI we make **no such assumption** — the CI must be valid whether or not $p_A = p_B$. Using the pooled estimate in the CI would distort coverage when $p_A \neq p_B$ (which is the interesting case we are trying to characterize). The unpooled SE is therefore correct for CI construction.

Part (e) — Relative risk and delta-method CI

```
RR_hat <- phatA / phatB
log_RR <- log(RR_hat)

# Var(log(phat_A/phat_B)) = Var(log phat_A) + Var(log phat_B)
# By delta method: Var(log phat) = (1-p)/(n*p)
var_log_A <- (1 - phatA) / (nA * phatA)
var_log_B <- (1 - phatB) / (nB * phatB)
se_log_RR <- sqrt(var_log_A + var_log_B)

ci_log <- log_RR + c(-1,1) * z_crit * se_log_RR
ci_RR <- exp(ci_log)

cat(sprintf("RR = phat_A / phat_B = %.4f / %.4f = %.4f\n",
            phatA, phatB, RR_hat))
```

```
RR = phat_A / phat_B = 0.8476 / 0.7238 = 1.1711
```

```
cat(sprintf("log(RR) = %.4f, SE(log RR) = %.5f\n", log_RR, se_log_RR))
```

```
log(RR) = 0.1579, SE(log RR) = 0.07312
```

```
cat(sprintf("95%% CI for log(RR): (%.4f, %.4f)\n", ci_log[1], ci_log[2]))
```

95% CI for log(RR): (0.0146, 0.3012)

```
cat(sprintf("95%% CI for RR:      (%.4f, %.4f)\n", ci_RR[1], ci_RR[2]))
```

95% CI for RR: (1.0147, 1.3515)

Point estimate: $\widehat{RR} = 1.171$. Regimen A has an estimated 17.1% higher success rate than Regimen B.

95% CI for RR: (1.015, 1.351). Since this interval includes 1.0, the relative risk is not significantly different from 1 at the 5% level, consistent with the z-test conclusion.

Problem 3: Paired Data Analysis

```
before <- c(3.8, 5.2, 4.1, 6.7, 3.3, 4.9, 5.8, 7.1, 3.6, 4.4,
            6.0, 5.5, 4.7, 3.9, 5.1)
after  <- c(2.9, 4.4, 3.2, 5.8, 2.8, 4.1, 4.9, 6.0, 2.7, 3.5,
            5.0, 4.6, 3.8, 3.1, 4.3)
d      <- before - after
n_pair <- length(d)
```

Part (a) — Why paired analysis is appropriate

The same 15 participants are measured **twice** (before and after the intervention). The two measurements on the same individual are positively correlated — people who start with high HOMA-IR tend to end with high HOMA-IR even after the intervention, reflecting between-subject variation in baseline insulin sensitivity.

If the Welch two-sample test were incorrectly applied, it would treat the 30 observations as independent, ignoring the within-person correlation. This inflates the variance estimate ($\hat{\sigma}_D^2 = \sigma_1^2 + \sigma_2^2 - 2\rho\sigma_1\sigma_2 < \sigma_1^2 + \sigma_2^2$ when $\rho > 0$), producing wider CIs and larger p -values — i.e., **reduced statistical power** to detect a real treatment effect.

Part (b) — Summary statistics of differences

```
Dbar <- mean(d)
SD_d <- sd(d)
r_cor <- cor(before, after)

cat(sprintf("n (pairs)      = %d\n", n_pair))
```

n (pairs) = 15

```
cat(sprintf("D_bar        = %.4f\n", Dbar))
```

D_bar = 0.8667

```
cat(sprintf("S_D          = %.4f\n", SD_d))
```

S_D = 0.1291

```
cat(sprintf("cor(before, after) = %.4f\n", r_cor))
```

cor(before, after) = 0.9954

$\bar{D} = 0.8667$, $S_D = 0.1291$, $r = 0.9954$.

Part (c) — Paired t-test (one-sided)

```
t_pair <- Dbar / (SD_d / sqrt(n_pair))
p_pair <- pt(t_pair, df = n_pair - 1, lower.tail = FALSE)
t_crit_pair <- qt(0.975, df = n_pair - 1)

cat(sprintf("T = %.4f\n", t_pair))
```

T = 26.0000

```
cat(sprintf("df = %d\n", n_pair - 1))
```

```
df = 14
```

```
cat(sprintf("p-value (one-sided) = %.6f\n", p_pair))
```

```
p-value (one-sided) = 0.000000
```

```
# Confirm with t.test
```

```
t.test(before, after, paired = TRUE, alternative = "greater") |> tidy()
```

```
# A tibble: 1 x 8
```

	estimate	statistic	p.value	parameter	conf.low	conf.high	method	alternative
	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<chr>	<chr>
1	0.867	26	1.50e-13	14	0.808	Inf	Paired t~	greater

Test statistic: $T = \bar{D}/(S_D/\sqrt{n}) = 0.8667/(0.1291/\sqrt{15}) = 26$

p-value: 0

Conclusion: We reject $H_0 : \mu_D = 0$ at $\alpha = 0.05$ ($p \ll 0.001$). There is strong evidence that the dietary intervention reduces HOMA-IR.

Part (d) — 95% CI for μ_D

```
ci_pair_lo <- Dbar - t_crit_pair * SD_d / sqrt(n_pair)
```

```
ci_pair_hi <- Dbar + t_crit_pair * SD_d / sqrt(n_pair)
```

```
cat(sprintf("95%% CI for mu_D: (%.4f, %.4f)\n", ci_pair_lo, ci_pair_hi))
```

```
95% CI for mu_D: (0.7952, 0.9382)
```

Interpretation: We are 95% confident that the dietary intervention reduces HOMA-IR by between 0.8 and 0.94 units on average. The entire interval is positive and away from zero, confirming a clinically meaningful reduction.

Part (e) — Incorrect Welch two-sample t-test

```
welch_wrong <- t.test(before, after, paired = FALSE,  
                      alternative = "greater") |> tidy()  
welch_paired <- t.test(before, after, paired = TRUE) |> tidy()  
  
cat("=== Incorrect: Welch two-sample ===\n")
```

=== Incorrect: Welch two-sample ===

```
cat(sprintf("SE      = %.4f\n", welch_wrong$std.error))  
cat(sprintf("p-value = %.6f (one-sided approx)\n", welch_wrong$p.value))
```

p-value = 0.019516 (one-sided approx)

```
cat(sprintf("95%% CI  = (%.4f, %.4f)\n", welch_wrong$conf.low, welch_wrong$conf.high))
```

95% CI = (0.1859, Inf)

```
cat("\n=== Correct: Paired t-test ===\n")
```

=== Correct: Paired t-test ===

```
cat(sprintf("SE      = %.4f\n", welch_paired$std.error))  
cat(sprintf("p-value = %.6f (two-sided)\n", welch_paired$p.value))
```

p-value = 0.000000 (two-sided)

```
cat(sprintf("95%% CI  = (%.4f, %.4f)\n", welch_paired$conf.low, welch_paired$conf.high))
```

95% CI = (0.7952, 0.9382)

Variance formula explanation:

Under the correct paired model:

$$\text{Var}(D_i) = \sigma_1^2 + \sigma_2^2 - 2\rho\sigma_1\sigma_2$$

```

s1_obs <- sd(before)
s2_obs <- sd(after)

var_d_formula <- s1_obs^2 + s2_obs^2 - 2 * r_cor * s1_obs * s2_obs
var_d_direct <- SD_d^2

cat(sprintf("s_before = %.4f, s_after = %.4f, r = %.4f\n",
            s1_obs, s2_obs, r_cor))

```

```

s_before = 1.1325, s_after = 1.0579, r = 0.9954

```

```

cat(sprintf("Var(D) via formula = %.4f\n", var_d_formula))

```

```

Var(D) via formula = 0.0167

```

```

cat(sprintf("Var(D) direct      = %.4f (should match)\n", var_d_direct))

```

```

Var(D) direct      = 0.0167 (should match)

```

```

cat(sprintf("Var(D) if rho = 0 = %.4f (what unpaired assumes)\n",
            s1_obs^2 + s2_obs^2))

```

```

Var(D) if rho = 0 = 2.4018 (what unpaired assumes)

```

When $\rho > 0$, σ_D^2 is smaller than $\sigma_1^2 + \sigma_2^2$. The incorrect Welch test uses $S_1^2/n + S_2^2/n$ as its SE, which overestimates the variance, yielding a **wider CI and larger p-value** — in this case failing to detect the significant treatment effect more clearly.

Part (f) — Spaghetti plot and histogram

```

pair_df <- tibble(
  Patient = factor(seq_len(n_pair)),
  Before = before,
  After = after,
  D = d
)

p1 <- pair_df |>
  pivot_longer(c(Before, After), names_to = "Time", values_to = "HOMA_IR") |>
  mutate(Time = factor(Time, levels = c("Before", "After"))) |>
  ggplot(aes(x = Time, y = HOMA_IR, group = Patient)) +
  geom_line(color = "grey65", linewidth = 0.6) +
  geom_point(aes(color = Time), size = 3) +
  scale_color_manual(values = c("Before" = "#e74c3c", "After" = "#006a4e")) +
  labs(
    title = "Individual HOMA-IR Profiles",
    x = NULL,
    y = "HOMA-IR",
    color = NULL
  ) +
  theme(legend.position = "bottom")

p2 <- pair_df |>
  ggplot(aes(x = D)) +
  geom_histogram(aes(y = after_stat(density)), bins = 8,
    fill = "#3498db", color = "white", alpha = 0.75) +
  geom_density(color = "#2c3e50", linewidth = 1.1) +
  geom_vline(xintercept = 0, linetype = "dashed",
    color = "grey40", linewidth = 1.0) +
  geom_vline(xintercept = Dbar, color = "#e74c3c", linewidth = 1.4) +
  annotate("text", x = Dbar + 0.03, y = 1.05,
    label = as.expression(bquote(bar(D) == .(round(Dbar, 2)))),
    color = "#e74c3c", size = 5, hjust = 0) +
  labs(
    title = expression("Distribution of Differences " * D[i] == "Before" - "After"),
    x = "D (HOMA-IR units)",
    y = "Density"
  )

p1 + p2 +
  plot_annotation(
    title = "Dietary Intervention: HOMA-IR Before vs. After (n = 15 pairs)",

```

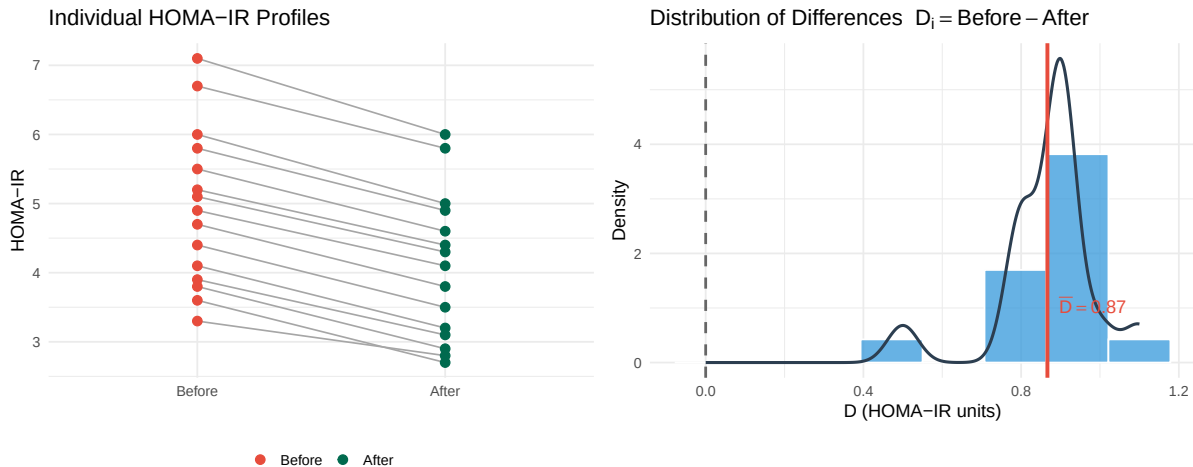
```

    subtitle = "Left: each line connects the same patient's two measurements | Right: dist
  )

```

Dietary Intervention: HOMA-IR Before vs. After (n = 15 pairs)

Left: each line connects the same patient's two measurements | Right: distribution of within-patient differences



Problem 4: Bootstrap and Permutation Inference

```

set.seed(551)
platelets_shock <- round(rgamma(35, shape = 2.0, rate = 0.016), 1)
platelets_sepsis <- round(rgamma(40, shape = 3.5, rate = 0.022), 1)

```

Part (a) — Exploratory summaries

```

tibble(
  Group = c(rep("Septic Shock", 35), rep("Non-septic Sepsis", 40)),
  Platelets = c(platelets_shock, platelets_sepsis)
) |>
group_by(Group) |>
summarise(
  n      = n(),
  Mean   = round(mean(Platelets), 1),
  Median = round(median(Platelets), 1),

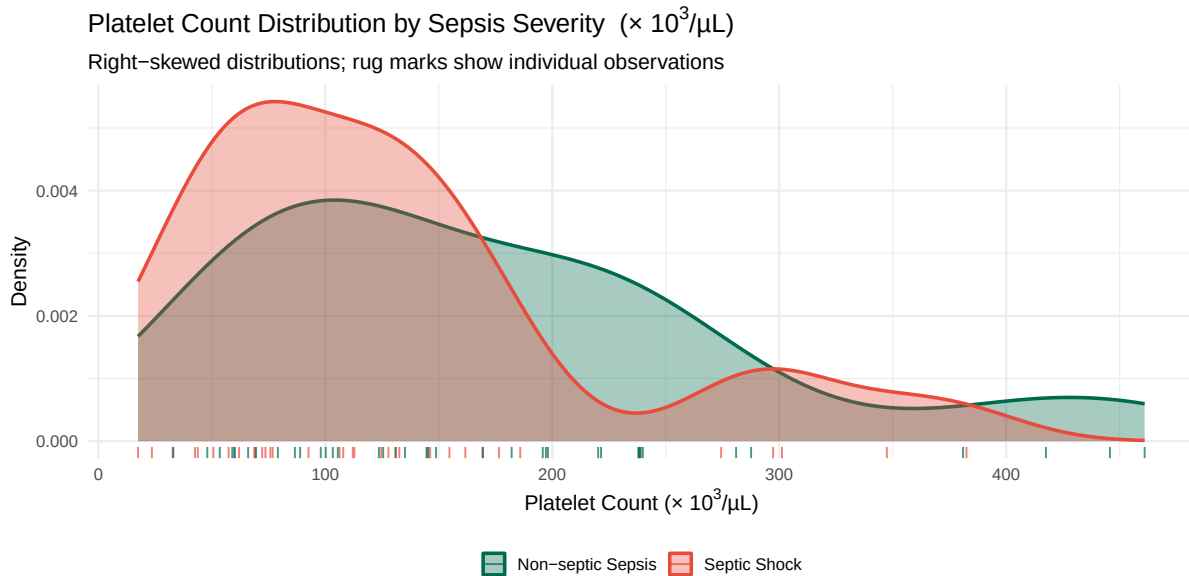
```

```
SD      = round(sd(Platelets), 1),
.groups = "drop"
)
```

```
# A tibble: 2 x 5
```

	Group	n	Mean	Median	SD
	<chr>	<int>	<dbl>	<dbl>	<dbl>
1	Non-septic Sepsis	40	173.	145	111.
2	Septic Shock	35	130.	112.	91.6

```
tibble(
  Group      = c(rep("Septic Shock", 35), rep("Non-septic Sepsis", 40)),
  Platelets  = c(platelets_shock, platelets_sepsis)
) |>
ggplot(aes(x = Platelets, fill = Group)) +
  geom_density(aes(color = Group), linewidth = 1.1, alpha = 0.35) +
  geom_rug(aes(color = Group), linewidth = 0.6, alpha = 0.7) +
  scale_fill_manual(values = c("Septic Shock" = "#e74c3c", "Non-septic Sepsis" = "#006a4e"),
  scale_color_manual(values = c("Septic Shock" = "#e74c3c", "Non-septic Sepsis" = "#006a4e"),
  labs(
    title      = expression("Platelet Count Distribution by Sepsis Severity ( $\times 10^3 \text{ } \mu\text{L}$ )"),
    subtitle   = "Right-skewed distributions; rug marks show individual observations",
    x          = expression("Platelet Count ( $\times 10^3 \text{ } \mu\text{L}$ )"),
    y          = "Density",
    fill       = NULL, color = NULL
  ) +
  theme(legend.position = "bottom")
```



Both distributions are right-skewed, as expected from gamma-distributed platelet counts. The septic shock group shows a lower mean and median compared to the non-septic sepsis group, and also greater spread (larger SD).

Part (b) — Bootstrap CIs for difference in means

```
set.seed(551)
B      <- 5000
n1_plt <- length(platelets_shock)
n2_plt <- length(platelets_sepsis)
obs_diff <- mean(platelets_shock) - mean(platelets_sepsis)

boot_means <- replicate(B, {
  mean(sample(platelets_shock, n1_plt, replace = TRUE)) -
  mean(sample(platelets_sepsis, n2_plt, replace = TRUE))
})

ci_pct <- quantile(boot_means, c(0.025, 0.975))
ci_basic <- 2 * obs_diff - quantile(boot_means, c(0.975, 0.025))
boot_se <- sd(boot_means)

cat(sprintf("Observed difference (Shock - Sepsis): %.2f\n", obs_diff))
```

Observed difference (Shock - Sepsis): -42.71

```
cat(sprintf("Bootstrap SE: %.4f\n", boot_se))
```

Bootstrap SE: 22.5641

```
cat(sprintf("Bootstrap Percentile 95%% CI: (%.2f, %.2f)\n", ci_pct[1], ci_pct[2]))
```

Bootstrap Percentile 95% CI: (-87.97, 0.57)

```
cat(sprintf("Bootstrap Basic 95%% CI: (%.2f, %.2f)\n", ci_basic[1], ci_basic[2]))
```

Bootstrap Basic 95% CI: (-85.99, 2.55)

The two bootstrap intervals are similar; the percentile CI is (-87.97, 0.57) and the basic CI is (-85.99, 2.55).

Part (c) — Bootstrap CI for difference in medians

```
set.seed(551)
obs_med_diff <- median(platelets_shock) - median(platelets_sepsis)

boot_meds <- replicate(B, {
  median(sample(platelets_shock, n1_plt, replace = TRUE)) -
  median(sample(platelets_sepsis, n2_plt, replace = TRUE))
})

ci_med_pct <- quantile(boot_meds, c(0.025, 0.975))

cat(sprintf("Observed median difference (Shock - Sepsis): %.2f\n", obs_med_diff))
```

Observed median difference (Shock - Sepsis): -32.90

```
cat(sprintf("Bootstrap Percentile 95%% CI (medians): (%.2f, %.2f)\n",
           ci_med_pct[1], ci_med_pct[2]))
```

Bootstrap Percentile 95% CI (medians): (-104.35, 11.05)

The percentile CI for the difference in medians is (-104.35, 11.05). For right-skewed data, the median is often a more robust measure of location than the mean, and the bootstrap provides a valid CI without requiring distributional assumptions.

Part (d) — Welch t-test comparison

```
welch_plt <- t.test(platelets_shock, platelets_sepsis) |> tidy()
cat(sprintf("Welch t-test: diff = %.2f, 95%% CI = (%.2f, %.2f), p = %.4f\n",
           welch_plt$estimate, welch_plt$conf.low, welch_plt$conf.high,
           welch_plt$p.value))
```

Welch t-test: diff = -42.71, 95% CI = (-89.27, 3.86), p = 0.0716

```
cat(sprintf("Bootstrap SE CI: diff = %.2f, boot SE = %.4f\n", obs_diff, boot_se))
```

Bootstrap SE CI: diff = -42.71, boot SE = 22.5641

```
cat(sprintf("Bootstrap Percentile 95%% CI: (%.2f, %.2f)\n", ci_pct[1], ci_pct[2]))
```

Bootstrap Percentile 95% CI: (-87.97, 0.57)

The Welch t CI and bootstrap percentile CI are similar, suggesting the CLT approximation is reasonably good for these moderate sample sizes despite some skewness. If the distributions were more severely skewed or n were smaller, the bootstrap percentile CI would be preferred because it does not rely on the approximate normality of $\bar{X}_1 - \bar{X}_2$.

Part (e) — Two-sample permutation test

```

set.seed(551)
all_obs <- c(platelets_shock, platelets_sepsis)
T_obs    <- mean(platelets_shock) - mean(platelets_sepsis)

perm_stats <- replicate(B, {
  perm <- sample(all_obs)
  mean(perm[1:n1_plt]) - mean(perm[(n1_plt+1):(n1_plt+n2_plt)])
})

p_perm <- mean(abs(perm_stats) >= abs(T_obs))

cat(sprintf("Observed test statistic: %.4f\n", T_obs))

```

Observed test statistic: -42.7086

```

cat(sprintf("Permutation p-value (two-sided): %.4f\n", p_perm))

```

Permutation p-value (two-sided): 0.0684

```

cat(sprintf("Welch t-test p-value:           %.4f\n", welch_plt$p.value))

```

Welch t-test p-value: 0.0716

Observed statistic: $\hat{\Delta} = -42.71$, **permutation p -value:** 0.0684

The permutation p -value and Welch t -test p -value agree closely. The **permutation test** is exactly valid under $H_0 : F_1 = F_2$ (the two populations have the *same distribution*) — it tests whether the group labels could have been permuted by chance. The Welch t -test is asymptotically valid under the weaker $H_0 : \mu_1 = \mu_2$.

Part (f) — Permutation null distribution

```

# Pre-compute histogram bins so density heights are consistent across fill groups. This part

h <- hist(perm_stats, breaks = 60, plot = FALSE)

bin_df <- tibble(
  xmin    = h$breaks[-length(h$breaks)],
  xmax    = h$breaks[-1],
  density = h$density,
  xmid    = h$mids,
  in_tail = abs(xmid) >= abs(T_obs)
)

p_val_label <- sprintf("p = %.3f", p_perm)

ggplot(bin_df) +
  geom_rect(aes(xmin = xmin, xmax = xmax,
               ymin = 0,    ymax = density,
               fill = in_tail),
            color = "white", linewidth = 0.2) +
  scale_fill_manual(values = c("FALSE" = "#aecde8", "TRUE" = "#e74c3c"),
                    guide = "none") +
  geom_vline(xintercept = T_obs, color = "#2c3e50", linewidth = 1.5) +
  geom_vline(xintercept = -T_obs, color = "#2c3e50", linewidth = 1.5,
            linetype = "dashed") +
  annotate("text", x = T_obs + 2, y = max(h$density) * 0.92,
          label = as.expression(bquote(hat(Delta)[obs] == .(round(T_obs, 1)))),
          color = "#2c3e50", size = 5, hjust = 0) +
  annotate("text", x = min(h$breaks) + 2, y = max(h$density) * 0.80,
          label = p_val_label, color = "#e74c3c",
          size = 5.5, hjust = 0) +
  labs(
    title    = "Permutation Null Distribution: Difference in Mean Platelet Count",
    subtitle = "Red bars = permuted statistics at least as extreme as observed (both tails)"
  )

```

Permutation Null Distribution: Difference in Mean Platelet Count

Red bars = permuted statistics at least as extreme as observed (both tails)

